

① (a) Define Data mining:-

Data mining is the process of discovering useful patterns, relationships, and knowledges from large amounts of the data.

Two Basic Data mining Tasks:-

① classification

② clustering

Steps in KDD:-

① Data cleaning

② Data integration

③ Data selection

④ Data transformation

⑤ Data mining

⑥ Pattern Evaluation

⑦ Knowledge presentation

(b) Data cleaning Data cleaning is the process of detecting and correcting inaccurate, incomplete, or inconsistent data.

Two data quality issues:-

① missing values

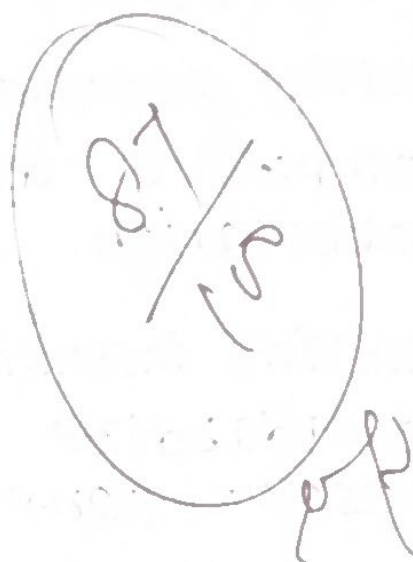
② noisy data

Handling missing values:-

→ Replace with mean / median / mode

Ex:- missing Age = replace with average age.

incorrect value = smooth using binning.



2 @ data integration:- data integration combines data from multiple sources into one unified dataset.

challenges:-

① Data redundancy

② Data inconsistency.

Handling redundancy:-

-> Improve duplicate records.

-> correlation analysis.

Handling inconsistency:-

-> standardize formats

-> resolve conflicting values

6)

covariance Analysis:-

covariance analysis measures how two variables change together.

two covariance forms:-

① positive covariance.

② negative covariance.

③ a) Given

$A = 2, 3, 5, 4, 1$

$B = 5, 8, 10, 11, 14$

Both A and B generally increase together

The Covariance is Positive, so the stock prices rise together, not independently.

⑥ Architecture of Data mining system:-

Data sources

Data base / Data warehouse.

Data cleaning & integration.

Data mining Engine

Pattern Evaluation.

Knowledge Base.

User Interface.

Components:-

- > Data base / Data warehouse - stores data.
- > Data cleaning - removes errors.
- > Data mining Engine - finds patterns.
- > Pattern Evaluation - selects useful pattern.

④ a) Given

$$x = \{4, 5, 6, 7, 9\} \quad y = \{8, 3, 7, 5, 6\}$$

$$\text{mean}(x) = 6.2$$

$$\text{mean}(y) = 5.8$$

$$\text{COV}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n}$$

x	y	$(x - 6.2)$	$(y - 5.8)$	Product
4	8	-2.2	2.2	-4.84
5	3	-1.2	-2.8	3.36
6	7	-0.2	1.2	-0.24
7	5	0.8	-0.8	-0.64
9	6	2.8	0.2	0.56

$$\text{sum} = -1.80$$

$$\text{covariance}$$

$$= -1.80/5$$

$$= -0.36$$

(b) major steps of data preprocessing:-
 ① Data cleaning ② Data integration ③ Data transformation ④ Data reduction ⑤ Data discretization

Pipeline:-

Missing values $\rightarrow$ Remove outliers $\rightarrow$ Standardize
 Formats $\rightarrow$ Normalize Data $\rightarrow$ Reduce Data $\rightarrow$
 Discretize Data.

⑤ (a)

Data set:

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33,
 35, 35, 35, 36, 40, 45, 46, 52, 70

Equal frequency

Bin 1: 13, 15, 16, 16, 19, 20, 20, 21, 22

mean = 17.

Smoothed: 17, 17, 17, 17, 17, 17, 17, 17

Bin 2: 22, 25, 25, 25, 25, 30, 33, 33, 35

mean = 28.

Smoothed: 28, 28, 28, 28, 28, 28, 28, 28

Bin 3: 35, 35, 35, 36, 40, 45, 46, 52, 70.

mean = 43

Smoothed: 43, 43, 43, 43, 43, 43, 43, 43

b)

data discretization it converts continuous values into intervals (bins).

Importance:-

- > Reduces data size.
- > Removes noise
- > Improves mining accuracy.

Equal width

① Equal interval size

② Simple to implement

③ may create empty bins

Equal frequency

① Equal number of records

② Better for uneven data.

③ Balanced distribution.

Ex:-

Equal width : 0-20, 21-40, 41-60

Equal frequency : Each bin contains the same number of values.